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Project 1

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1.0 Introduction

For this assignment, we are tasked to collect and analyse the dataset chose to identify the potential business insights. We collected a dataset that represents the customer's shopping behaviour in the fashion industry. It provides comprehensive information about customer's product preferences and spending habits during their shopping experience. This dataset encompasses a range of variables, from customers profiling and clothing categories to payment details. We hope that through this dataset, we can clearly showcase the customers' shopping preferences with data visualization, thus aiding the company in identifying the market trends to increase the business sales and profits.

1.1 Overview of the dataset

The dataset mainly focuses on the customers' spending behaviour in the fashion industry. The dataset is sourced from Kaggle, a data science platform which contains various types of datasets. It was first published by Shahzad Aslam and the dataset is scheduled to be updated annually. This dataset contains 3900 rows and 21 columns. The dataset is fully completed, and no null or duplicate data are detected. Below is the list of data that are included in this dataset:

1. Customer ID	6. Purchase Amount (USD)	11. Review Rating	16. Previous Purchases
2. Age	7. Location	12. Subscription Status	17. Payment Method
3. Gender	8. Size	13. Shipping Type	18. Frequency of Purchases
4. Item Purchased	9. Colour	14. Discount Applied	19. Discount Percentage
5. Category	10. Season	15. Promo Code Used	20. Customer Income Level
			21. Item Price

Table 1.1.1: Data Set List

1.2 Objective of the analysis

The primary goal of this analysis is to employ classification models to understand and predict customer behaviour patterns. Specifically, the analysis aims to:

1. **Classify customers based on age group dominance:** Identifying which age groups are more prevalent in the customer base and understanding their characteristics.
2. **Determine payment method preferences:** Classifying customers based on their chosen method of payment and identifying demographic trends.
3. **Assess the influence of income levels on purchasing behaviour:** Investigating whether income brackets significantly influence how often or how much customers spend.

By building and evaluating multiple classification models, the analysis will reveal underlying trends and relationships that can inform marketing strategies, product placement, and customer experience optimization.

1.3 Business insights to be explored

Understanding customer behaviour is critical to developing data-driven business strategies. The analysis will aim to extract the following actionable insights:

1. **Dominant Customer Demographics**: Identifying which age groups form the core of the customer base can help tailor products and marketing strategies more effectively.

2. **Preferred Payment Channels:** Understanding payment preferences can guide financial partnerships (e.g., with digital wallet providers), streamline checkout processes, and increase customer satisfaction.

3. **Income vs. Spending Behaviour:** Analysing the relationship between income and purchasing frequency or volume will help in segmenting customers by value and targeting high-value segments with loyalty programs or premium offerings.

These insights are crucial for businesses seeking to improve customer acquisition, retention, and overall satisfaction by aligning operational strategies with customer preferences and behaviours.

1.4 Classification models selected for the analysis

To perform the classification tasks outlined above, the following three machine learning models were selected due to their simplicity, interpretability, and effectiveness in handling categorical data.

1.4.1 K-Nearest Neighbours (K-NN)

K-NN is a non-parametric algorithm that classifies data points based on the class of their nearest neighbours in the feature space. It does not make any assumptions about the underlying data distribution and works well with both numerical and categorical data. Its simplicity makes it a good baseline model.

Advantages: Easy to implement, interpretable, requires no training phase.

Limitations: Sensitive to irrelevant features and requires appropriate distance metric selection.

1.4.2 Naïve Bayes

Naïve Bayes is a probabilistic classifier based on Bayes' theorem, assuming independence among predictors. Despite its simplicity, it performs surprisingly well in real-world scenarios involving categorical input features.

Advantages: Fast, works well with high-dimensional data, suitable for multi-class classification.

Limitations: Assumes independence among features, which may not hold true in all cases.

1.4.3 Decision Tree

A decision tree builds a model in the form of a tree structure, using branches to represent decision rules and leaves to represent outcomes. It is highly interpretable and can handle both categorical and numerical data without the need for normalization.

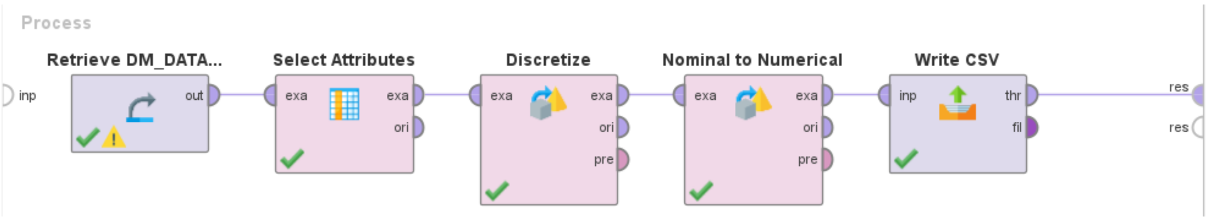
Advantages: Interpretable, handles non-linear relationships, requires little data preprocessing.

Limitations: Prone to overfitting if not properly pruned.

These models will be trained and evaluated across multiple classification targets to determine which performs best for different types of business questions. Their performance will be compared using metrics such as accuracy, precision, recall, and F1-score in Section 4.

2.0 Data Preparation

Before conducting the machine learning analysis, we had done the data preprocessing to ensure the dataset is cleaned and prepared for reliable results. The Data preparation was performed using **RapidMiner**.



2.1 Attribute Filtering

We use Select Attributes operator remove the Customer ID column from the dataset. This column was excluded because it only serves as a unique identifier and does not contain any meaningful information for data analysis. All other columns were kept for further processing.

2.2 Data Categorization

The Discretize by User Specification operator used to convert the continuous Age attribute into categorical bins. This process simplifies the analysis by grouping numerical values into labelled intervals, making it easier to handle categorical representations in later steps.

2.3 Data Transformation

The Nominal to Numerical operator used to convert all nominal (text-based) attributes into numeric form using unique integer encoding. This conversion is essential for compatibility with most machine learning algorithms, which typically require numerical input. Each unique category was automatically assigned a specific number.

2.4 Data Export

The Write CSV operator used to export the finalized dataset into a new CSV file. This output file contains all selected and transformed attributes in a structured format, including headers and comma-separated values, making it suitable for subsequent model training and evaluation.

2.5 Sampling strategy for model training and testing

In this study, the dataset is split into 70% for training and 30% for testing to ensure the model is trained on a sufficient amount of data while maintaining a separate subset for unbiased evaluation. The training data is used to teach the model the relationship between features and the target variable, while the testing data is used to assess the model's generalization ability.

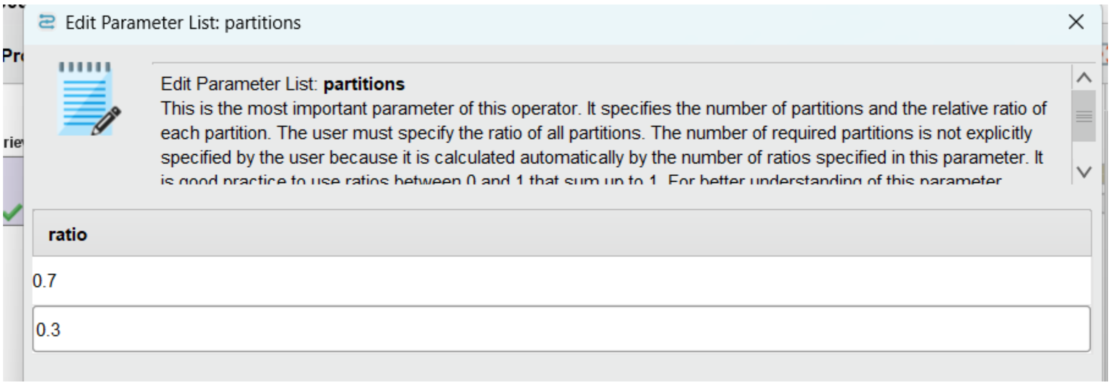


Figure 2.5.1 Split Data Operator in RapidMiner

3.0 Classification Models

This section assesses three machine learning models to get actionable insights from consumer data in the fashion industry: KNN, Naïve Bayes, and Decision Tree. The models' performance is evaluated by precision, recall, F1-score, and accuracy. A **70:30** split of the pre-processed dataset is used for training and testing.

3.1 Age Group Dominance

Age is a crucial demographic factor that influences purchasing behaviour. The dataset's "Age" column offers an insightful view into how different age groups engage with online shopping. By analysing the data, we can observe distinct trends in how age impacts customer purchasing behaviour and item selection.

3.1.1 K-Nearest Neighbours (K-NN)

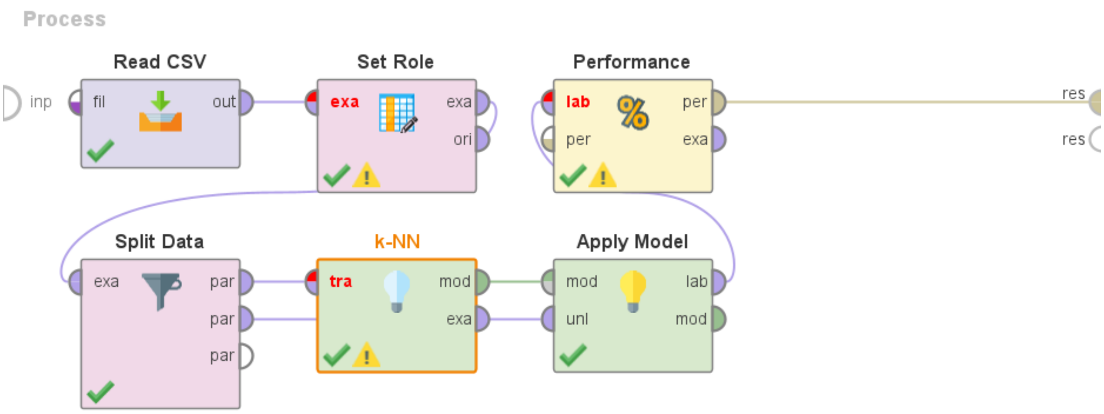


Figure 3.1.1.1 Dataset trained using K-Nearest Neighbours Classification Model

accuracy: 38.63%

	true Adult	true Young Adult	true Senior	true Teen	class precision
pred. Adult	318	213	132	8	47.39%
pred. Young Adult	170	115	66	9	31.94%
pred. Senior	70	45	19	4	13.77%
pred. Teen	1	0	0	0	0.00%
class recall	56.89%	30.83%	8.76%	0.00%	

Figure 3.1.1.2 Result of K-Nearest Neighbour Classification Model

In the KNN classification model for 3.1 Age Group Dominance, the overall accuracy is 38.63%, indicating that the model correctly predicts the age group in about one-third of the cases. Precision measures the proportion of correct predictions for each age group out of all predictions made for that group. For instance, 'Adult' has the highest precision (47.39%), meaning nearly half of the instances predicted as 'Adult' were actually 'Adult', while 'Teen' has

0% precision, showing complete misclassification. Recall reflects how well the model identifies all instances of a given class. The model recalls 56.89% of actual 'Adults' but performs poorly for 'Teen' (0%) and 'Senior' (8.76%), indicating that it fails to recognize most instances of those groups. The F1 score, which balances precision and recall, would be especially low for underperforming classes like 'Teen' and 'Senior', highlighting the model's ineffectiveness at handling imbalanced or overlapping class distributions in this task.

3.1.2 Naïve Bayes

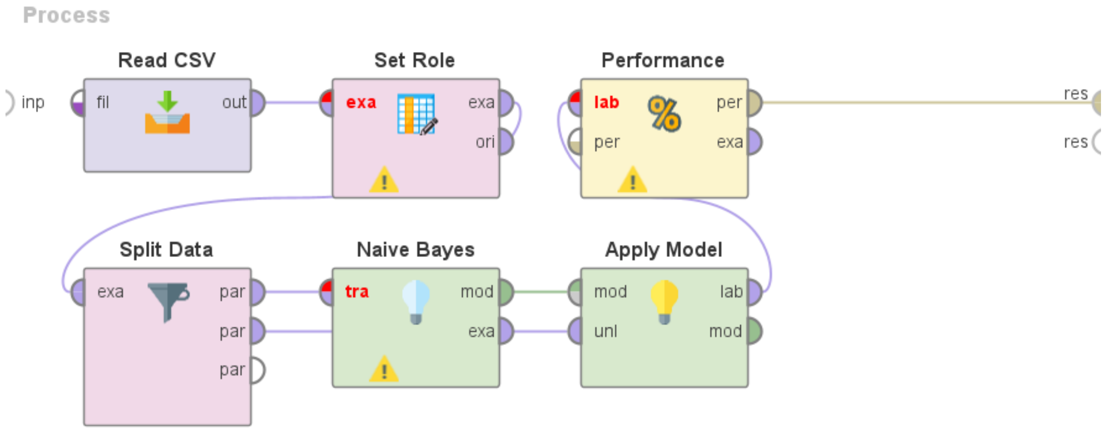


Figure 3.1.2.1 Dataset trained using Naïve Bayes Classification Model

accuracy: 31.97%

	true Adult	true Young Adult	true Senior	true Teen	class precision
pred. Adult	4	1	0	0	80.00%
pred. Young Adult	22	159	3	11	81.54%
pred. Senior	523	139	206	5	23.60%
pred. Teen	10	74	8	5	5.15%
class recall	0.72%	42.63%	94.93%	23.81%	

Figure 3.1.2.2 Result of Naïve Bayes Classification Model

In this Naïve Bayes classification model for 3.1 Age Group Dominance, the overall accuracy is 31.97%, indicating the model correctly classifies about one-third of the instances. The precision is highest for 'Young Adult' (81.54%) and 'Adult' (80.00%), meaning that when the model predicts those categories, it is usually correct. However, it performs poorly for 'Senior' (23.60%) and very poorly for 'Teen' (5.15%), indicating many false positives in those predictions. On the other hand, recall is exceptionally high for 'Senior' (94.93%), suggesting the model is very effective at identifying most actual 'Senior' instances. In contrast, recall is extremely low for 'Adult' (0.72%) and weak for 'Teen' (23.81%), meaning it fails to capture the majority of those actual cases. The F1 score, which balances both precision and recall, would be strong only for 'Senior', while it would be very low for 'Adult' and 'Teen', indicating the model's heavy bias toward recognizing one dominant class and its struggle with underrepresented or overlapping categories.

3.1.3 Decision Tree

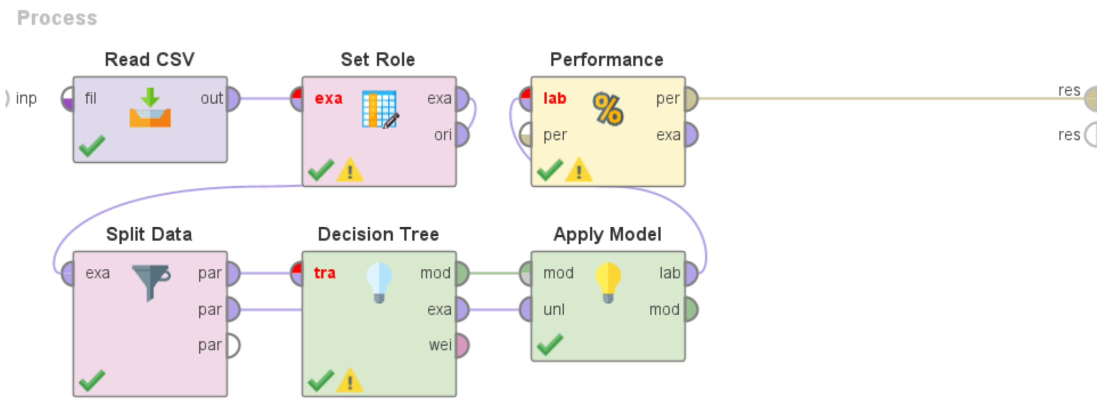


Figure 3.1.3.1 Dataset trained using Decision Tree Classification Model

Table View Plot View

accuracy: 58.63%

	true Adult	true Young Adult	true Senior	true Teen	class precision
pred. Adult	416	91	155	4	62.46%
pred. Young Adult	76	245	38	16	65.33%
pred. Senior	67	31	24	0	19.67%
pred. Teen	0	6	0	1	14.29%
class recall	74.42%	65.68%	11.06%	4.76%	

Figure 3.1.3.2 Result of Decision Tree Classification Model

The model achieved an overall accuracy of 58.63%, with varying performance across the different age groups. The "Adult" and "Young Adult" classes showed relatively strong results, with precision scores of 62.46% and 65.33%, and recall scores of 74.42% and 65.68% respectively, indicating that the model was both accurate and consistent in predicting these categories. In contrast, the "Senior" and "Teen" classes demonstrated significantly lower performance, with precision scores of just 19.67% and 14.29%, and recall scores of 11.06% and 4.76%, reflecting the model's difficulty in correctly identifying these underrepresented groups. Consequently, their F1 scores—representing the balance between precision and recall—are also very low, suggesting poor predictive power for these classes. Overall, while the model performs reasonably well on the dominant classes, it struggles with minority classes, likely due to class imbalance, which negatively impacts its generalization and fairness.

3.2 Payment Method Preferences

Payment method is important transactional component that affects how people buy. The dataset's "Payment Method" column provides a useful perspective on how various payment methods impact consumer buying habits. We can see clear patterns in the relationship between chosen payment methods and item selection, order value, and purchase frequency by studying the data.

3.2.1 K-Nearest Neighbours (K-NN)

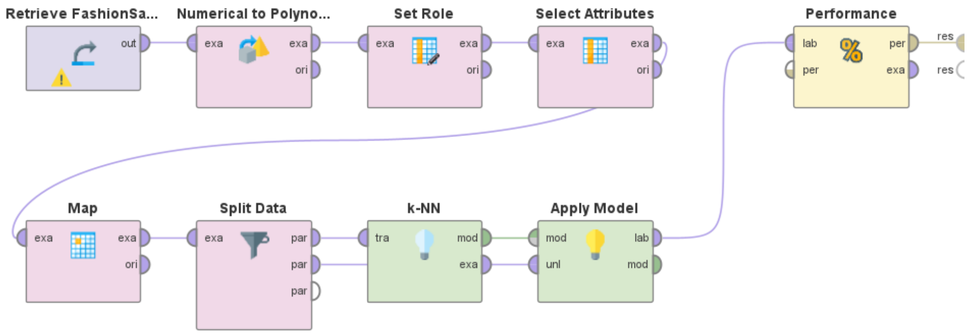


Figure 3.2.1.1 Dataset trained using K-Nearest Neighbour Classification Model

accuracy: 18.46%

	true Venmo	true Cash	true Credit Card	true Paypal	true Bank Transfer	true Debit Card	class precision
pred. Venmo	30	29	25	29	25	24	18.52%
pred. Cash	43	37	53	33	38	28	15.95%
pred. Credit Card	35	24	32	32	34	39	16.33%
pred. Paypal	44	52	41	64	42	49	21.92%
pred. Bank Transfer	16	35	22	26	25	23	17.01%
pred. Debit Card	22	24	28	19	20	28	19.86%
class recall	15.79%	18.41%	15.92%	31.53%	13.59%	14.66%	

Figure 3.2.1.2 Result of K-Nearest Neighbour Classification Model

The model's overall accuracy is 18.46%, with PayPal being the best-predicted class (31.53% recall, 21.92% precision). Venmo (15.79% recall) and Bank Transfers (13.59% recall) had the worst performance. All classes experienced broad misclassification, with false positives scattered rather evenly across categories (e.g., 24-29 Venmo misclassifications per projected class), despite the moderate precision (19.86%) displayed by debit cards. Low recall rates below 32% for all payment methods show that the model currently lacks solid predictive capacity overall, but showing marginally better distinction than earlier iterations, especially for PayPal transactions (64 true positives).

3.2.2 Naïve Bayes

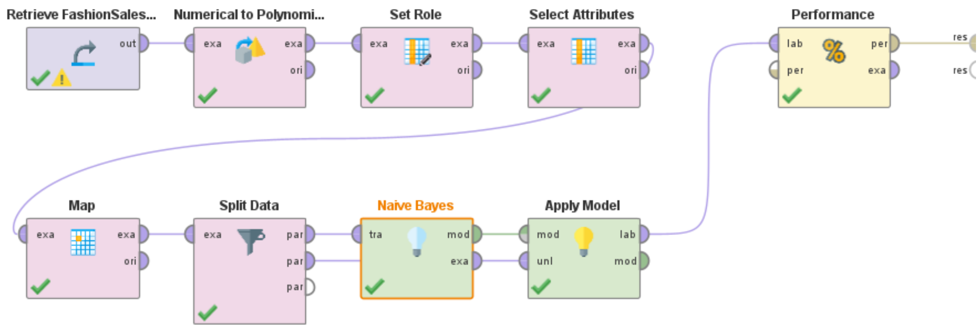


Figure 3.2.2.1 Dataset trained using Naïve Bayes Classification Model

accuracy: 16.84%

	true Venmo	true Cash	true Credit Card	true Paypal	true Bank Transfer	true Debit Card	class precision
pred. Venmo	20	19	27	15	24	17	16.39%
pred. Cash	28	29	27	34	28	30	16.48%
pred. Credit Card	33	44	33	33	39	35	15.21%
pred. Paypal	39	42	43	51	34	36	20.82%
pred. Bank Transfer	22	27	24	26	18	27	12.50%
pred. Debit Card	48	40	47	44	41	46	17.29%
class recall	10.53%	14.43%	16.42%	25.12%	9.78%	24.08%	

Figure 3.2.2.2 Result of Naïve Bayes Classification Model

The model achieved 16.84% overall accuracy with considerable categorisation problems across all payment methods. The model performed worst with bank transfer transactions (12.50% precision), but PayPal forecasts had the highest precision (20.82%). Venmo (10.53%) and bank transfers (9.78%) had the lowest recall rates, while PayPal (25.12%) and debit cards (24.08%) had somewhat higher recall rates. The model appears to lack substantial difference between payment modalities, as evidenced by the rather similar distribution of false positives across all predicted classes (e.g., 19-27 Venmo misclassifications per category). This suggests that no single method can be consistently detected. PayPal (51) and Debit Cards (46), which had the highest true positives, were still overshadowed by the large number of incorrect classifications in other categories.

3.2.3 Decision Tree

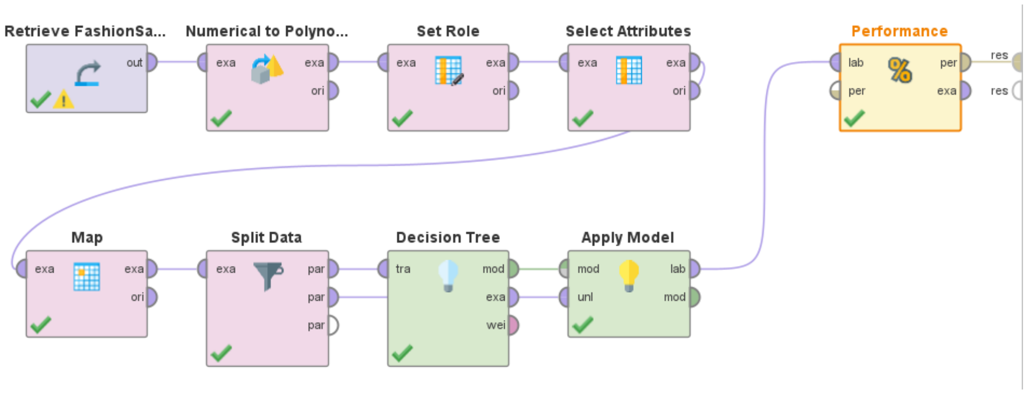


Figure 3.2.3.1 Dataset trained using Decision Tree Classification Model

accuracy: 16.84%

	true Venmo	true Cash	true Credit Card	true Paypal	true Bank Transfer	true Debit Card	class precision
pred. Venmo	3	0	2	1	1	1	37.50%
pred. Cash	176	182	182	190	171	171	16.98%
pred. Credit Card	7	9	5	6	7	9	11.63%
pred. Paypal	3	7	11	6	4	10	14.63%
pred. Bank Transfer	0	1	1	0	1	0	33.33%
pred. Debit Card	1	2	0	0	0	0	0.00%
class recall	1.58%	90.55%	2.49%	2.96%	0.54%	0.00%	

Figure 3.2.3.2 Result of Decision Tree Classification Model

Based on the confusion matrix, the model's overall accuracy is 16.84%, while its performance varied greatly throughout payment methods. Other approaches were often misclassified as Cash (171+ false positives), but 90.55% of genuine Cash transactions were successfully recognised (182 true positives). Minority groups showed especially poor performance, with only 3 Venmo (1.58% recall), 1 bank transfer (0.54% recall), and 0 debit card transactions (0% recall) accurately identified. While the model was able to forecast cash (16.98%) and bank transfers (33.33%) with moderate accuracy, it was unable to predict debit cards at all (0% precision). These findings imply that the algorithm struggles to identify minority payment classes and instead defaults to forecasting cash.

3.3 Impact of Customer Income Level on Purchasing Behaviour

Customer income level is crucial to determine purchasing behaviour. By analyzing how income levels relate to purchasing frequency, we can better understand consumer decision-making and predict purchasing patterns more effectively. This analysis focuses on the relationship between income and the frequency of purchases made by customers.

3.3.1 K-Nearest Neighbours (K-NN)

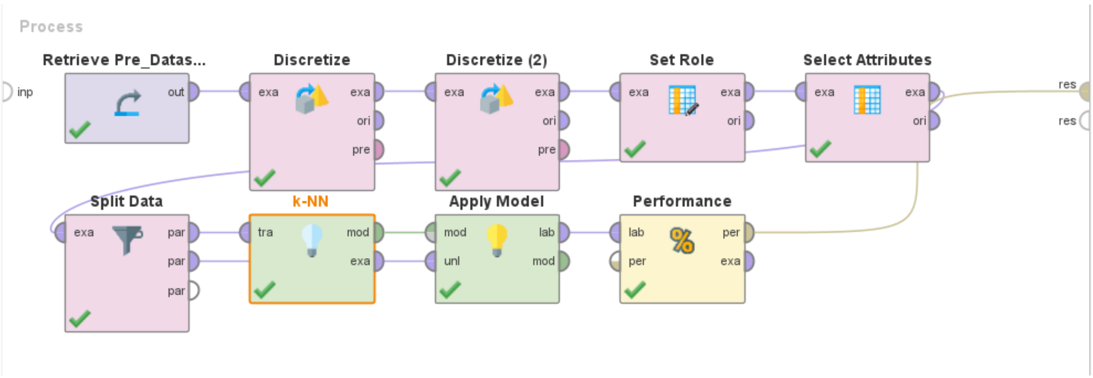


Figure 3.3.1.1 Dataset trained using K-NN Classification Model

accuracy: 34.87%

	true Low	true Medium	true High	class precision
pred. Low	237	188	160	40.51%
pred. Medium	131	75	85	25.77%
pred. High	128	70	96	32.65%
class recall	47.78%	22.52%	28.15%	

Figure 3.3.1.2 Result of K-NN Classification Model

In the KNN classification model for predicting Customer Income Level, the overall accuracy is 34.87%. For precision, which measures the accuracy of predictions for each income group shows that the Low income group has the highest precision at 40.51%, meaning that nearly 41% of the instances predicted as Low Income were actually Low Income. In contrast, the Medium Income group has a lower precision of 25.77%, while the High Income group has a precision of 32.65%, indicating the model's struggles with these classes. For, recall, which indicates how well the model identifies instances of each class, reveals that the model performs better in identifying Low Income customers, 47.78% but struggles significantly with Medium Income, 22.52% and High Income with 28.15% groups. This performance suggests that the model has difficulty with classifying the less frequent Medium and High Income classes, likely due to class imbalance.

3.3.2 Naïve Bayes

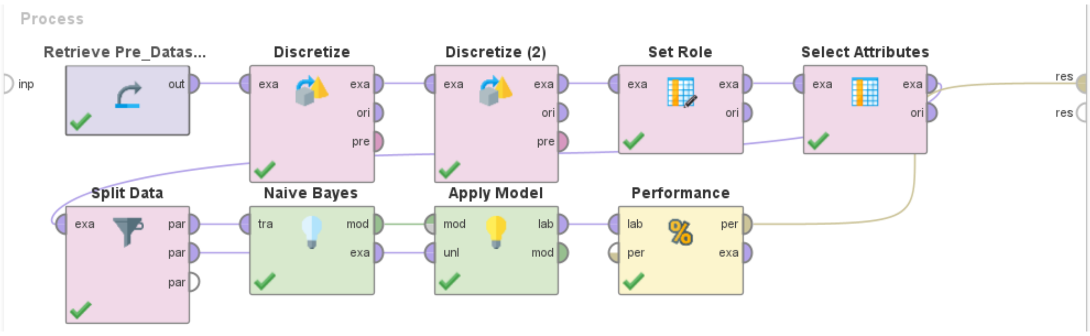


Figure 3.3.2.1 Dataset trained using Naïve Bayes Classification Model

accuracy: 38.72%

	true Low	true Medium	true High	class precision
pred. Low	361	243	259	41.83%
pred. Medium	64	41	31	30.15%
pred. High	71	49	51	29.82%
class recall	72.78%	12.31%	14.96%	

Figure 3.3.2.2 Result of Naïve Bayes Classification Model

In the Naïve Bayes classification model for predicting Customer Income Level, the overall accuracy is 38.72%. The precision value show that the Low Income group has the highest precision at 41.83%, showing that the model's predictions for Low Income are more accurate, while Medium Income and High Income have lower precision at 30.15% and 29.82%, respectively. The recall reflects that the model is able to correctly identify 72.78% of Low Income instances, but struggles with Medium Income, 12.31% and High Income, 14.96%. This performance suggests that while the model works better for Low Income predictions, it faces challenges in identifying Medium and High Income instances, it is possibly due to class imbalance or insufficient representation in the data.

3.3.3 Decision Tree

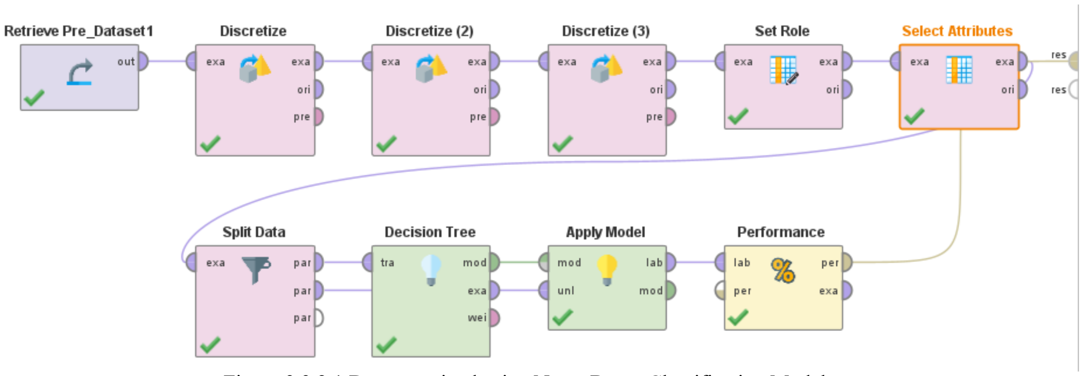


Figure 3.3.2.1 Dataset trained using Naïve Bayes Classification Model

accuracy: 34.24%

	true Medium	true Low	true High	class precision
pred. Medium	394	369	381	34.44%
pred. Low	0	2	3	40.00%
pred. High	11	6	5	22.73%
class recall	97.28%	0.53%	1.29%	

Figure 3.3.2.1 Result of Naïve Bayes Classification Model

In the Decision Tree classification model for predicting Customer Income Level, the overall accuracy is 34.24%. The precision value indicate that the Low Income group has a precision of 40.00%, meaning that when the model predicts Low Income, it is correct 40% of the time. The Medium Income group has a higher recall of 97.28%, indicating the model can identify most Medium Income instances, but its precision is much lower at 34.44%, showing that it misclassifies many instances as Medium Income. The High Income group has a recall of 1.29%, and a precision of 22.73%, indicating that the model performs poorly in identifying and classifying High Income customers. The low accuracy and the highly imbalanced recall across the classes suggest that the model is overfitting to the Medium Income group, likely due to class imbalance.

4.0 Model Evaluation

In this section, we observe, compare and evaluate each key business insights, mainly age group dominance, preferred payment methods and customers' income level. Our analysis is based on the performance metrics (precision, recall, F1-score and accuracy) from different classification models (K-Nearest Neighbour, Naïve Bayes and Decision Tree). Through this evaluation we can analyse which classification model best fit our business insights, allowing us to identify key variables to improve the fashion business sales.

4.1 Age Group Dominance

Age Group	KNN			Naïve Bayes			Decision Tree		
	P (%)	R (%)	F1(%)	P (%)	R (%)	F1(%)	P (%)	R (%)	F1(%)
Adult	47.39	56.89	51.68	80.00	0.72	1.43	62.46	74.42	67.92
Young Adult	31.94	30.83	31.37	81.54	42.63	56.01	65.33	65.68	65.50
Senior	13.77	8.76	10.70	23.60	94.93	37.80	19.67	11.06	14.16
Teen	0.00	0.00	NaN	5.15	23.81	8.47	14.29	4.76	7.14
Accuracy (%)	38.63			31.97			58.63		

P: Precision, **R:** Recall, **F1:** F1-score, **NaN:** Undefined

Table 4.1.1 Performance metrics of KNN, Naïve Bayes, and Decision Tree models across age groups

The Decision Tree model performs better on all important measures, making it the best classifier for predicting age groups. It maintains balanced precision-recall ratios for all age groups and achieves the highest overall accuracy (58.63%; see your previous correct table). It does especially well in the Adult (F1=67.92%) and Young Adult (F1=65.50%) categories. Despite its remarkable accuracy for young adults (81.54%), Naïve Bayes is unreliable for that crucial group due to its incredibly low recall for adults (0.72%). The KNN model performs moderately but consistently (F1-scores 31.37%-51.68%), however its overall efficacy is limited by its subpar performance for seniors and teens (F1=10.70%).

4.2 Payment Method

Payment Methods	KNN			Naïve Bayes			Decision Tree		
	P (%)	R (%)	F1(%)	P (%)	R (%)	F1(%)	P (%)	R (%)	F1(%)
Venmo	18.52	15.79	17.04	16.39	10.53	12.82	37.50	1.58	3.03
Cash	15.95	18.41	17.09	16.48	14.43	15.39	16.98	90.55	28.59
Credit Card	16.33	15.92	16.12	15.21	16.42	15.79	11.63	2.49	4.10
PayPal	21.92	31.53	25.86	20.82	25.12	22.77	14.63	2.96	4.92
Bank Transfer	17.01	13.59	15.11	12.50	9.78	10.97	33.33	0.54	1.06
Debit Card	19.86	14.66	16.87	17.29	24.08	20.13	0.00	0.00	NaN
Accuracy (%)	18.46			16.84			16.84		

P: Precision, **R:** Recall, **F1:** F1-score, **NaN:** Undefined

Table 4.1.2 Performance metrics of KNN, Naïve Bayes, and Decision Tree models across payment methods

With the highest overall accuracy (18.46%) out of the three models, the KNN model performs the best when it comes to predicting payment methods. KNN consistently maintains a moderate precision-recall balance across all payment types (F1-scores ranging from 15.11% to 25.86%), with notably excellent performance for PayPal transactions (F1=25.86%), even if no model exhibits exceptional results. Although it outperforms KNN for Debit Card predictions (F1=20.13% vs 16.87%), the Naïve Bayes classifier produces results that are comparable but marginally worse (accuracy=16.84%). The Decision Tree model shows severe recall imbalance thus proven mainly ineffective. It achieves 90.55% recall for cash payments but fails miserably on debit cards (NaN F1-score because of zero precision/recall). Although all models have trouble predicting the intricacy of payment methods, KNN is the most dependable option for business applications that need consistent, all-around performance rather than excelling in certain payment forms due to its best accuracy and consistent performance across categories.

4.3 Customer Income Level

Income Level	KNN			Naïve Bayes			Decision Tree		
	P (%)	R (%)	F1(%)	P (%)	R (%)	F1(%)	P (%)	R (%)	F1(%)
Low	40.51	47.78	43.86	41.83	72.78	53.13	34.44	97.28	50.87
Medium	25.77	22.52	24.04	30.15	12.31	17.48	40.00	0.53	1.05
High	32.65	28.15	30.23	29.82	14.96	19.92	22.73	1.29	2.44
Accuracy(%)	34.87			38.72			34.24		

P: Precision, **R:** Recall, **F1:** F1-score, **NaN:** Undefined

Table 4.1.3 Performance metrics of KNN, Naïve Bayes, and Decision Tree models across income levels

Out of the three models, the Naïve Bayes classifier performs the best overall for predicting income level, with the highest accuracy (38.72%). For low-income consumers, it performs noticeably better in recall (72.78%) and F1-score (53.13%), suggesting better identification of genuine low-income instances, even though its precision (41.83%) is marginally worse than KNN's (40.51%). Notably, Decision Tree's extremely low recall values (0.53%-1.29%) indicate serious under-detection, as evidenced by its poor performance for medium and high-income groups (F1-scores of 1.05% and 2.44%, respectively). Even though KNN exhibits balanced precision-recall tradeoffs across categories (F1-scores: 43.86%-24.04%-30.23%), Naïve Bayes is the best option for business applications where detecting lower-income clients is a top priority due to its greater accuracy and reliable low-income detection. Compared to the drastic fluctuations observed in Decision Tree results, the model's 17–53% F1-score range across categories also suggests more consistent performance.

5.0 Conclusion

In this project, we assessed three classification models: K-Nearest Neighbours (KNN), Naïve Bayes, and Decision Tree. The following were the main conclusions of the analysis, which concentrated on age group dominance, preferred payment methods, and income-level impact:

1. **Age Group Prediction**
   The Decision Tree model outperformed others (accuracy: 58.63%), excelling in identifying Adults (F1: 67.92%) and Young Adults (F1: 65.50%). Its balanced precision-recall ratios make it ideal for demographic targeting, though it struggled with underrepresented groups (Seniors/Teens).

2. **Payment Method Analysis**
   KNN achieved the highest accuracy (18.46%) and consistency across payment types, particularly for PayPal (F1: 25.86%). Decision Trees showed severe bias toward Cash (recall: 90.55%) but failed for Debit Cards (F1: NaN), highlighting limitations in handling imbalanced data.

3. **Income Level Classification**
   Naïve Bayes emerged as the best model (accuracy: 38.72%), with strong recall for Low-Income customers (72.78%) and balanced F1-scores (17–53%). Decision Trees, while capturing 97.28% of Low-Income cases, performed poorly for other groups (F1 < 2.5%), revealing overfitting.

Based on each model evaluation, we suggest a tiered method to convert these findings into workable strategies: Age-based marketing efforts should be driven by Decision Tree models for demographic targeting, especially for adults and young adults. In addition to focused efforts to improve data quality for underperforming categories like debit and credit cards, payment system optimisations should give priority to KNN's trustworthy PayPal transaction insights to improve digital payment integrations. Naïve Bayes' can be used to detect low-income buyers will be particularly helpful for customer segmentation, allowing for customised incentives and loyalty plans.

In conclusion, the Decision Tree (age groups), KNN (payment methods), and Naïve Bayes (income levels) offered the most trustworthy insights for their respective areas, even though no model consistently dominated all tasks. These findings enable data-driven choices to improve fashion industry sales tactics and consumer segmentation.

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